Neural-Symbolic Post-Tonal Composition Assistant with Pitch-Class Set, Serial, and Rhythmic-Profile Constraints

融合音级集合、序列与节奏轮廓约束的神经符号后调性作曲辅助方法

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Abstract

Explicit control of post-tonal material remains underrepresented in symbolic music generation, while evaluation on copyrighted contemporary repertoire is difficult to release and reproduce. We present a score-level neural-symbolic composition assistant conditioned on pitch-class sets and interval vectors, optional twelve-tone rows and serial transformations, rhythmic profiles, gestures, instrumentation, voice count, and score span. A legal procedural generator supplies 20,000 training, 2,000 validation, and 2,000 test fragments. A six-layer causal Transformer is trained on condition-preserving event windows, and symbolic analysis reranks sampled candidates before exporting MusicXML and a JSON report. On the fixed 2,000-condition test split, increasing the candidate budget from one to four and applying guided selection raised pc-set coverage from 0.9963 to 0.9999 and cyclic row-order accuracy from 0.1581 to 0.2411, while reducing interval-vector distance from 0.2390 to 0.0045 and rhythmic-profile distance from 0.5113 to 0.5041. Gesture consistency rose from 0.4772 to 0.4937. Removing each condition field from separately trained single-candidate models degraded its corresponding automatic endpoint. All 260 attempted exports across thirteen evaluated configurations passed structural, measure-count, and part-count checks, and twenty proposed-model examples were packaged with reports. However, strict serial-transformation accuracy remained zero for every neural configuration, and the proposed model's mean content-span ratio was 0.5195. The findings therefore concern automatic constraint adherence on a synthetic testbed, not artistic quality or stylistic authenticity.

Keywords: computer-assisted composition; post-tonal music; symbolic music generation; pitch-class set theory; twelve-tone serialism; constraint-guided decoding; MusicXML

1 Introduction

Post-tonal composition frequently begins from explicit symbolic materials: a pitch-class collection, an interval-class profile, a twelve-tone row and its transformations, a registral trajectory, or a characteristic rhythmic surface. Computer-assisted composition environments have long supported the manipulation of such structures, but recent neural music systems are more often evaluated on tonal continuation, popular-song structure, performance MIDI, or audio generation (Assayag et al., 1999; Briot et al., 2020). Those settings do not directly answer a practical question faced by composers of contemporary score-based music: can a generative model produce an editable sketch while preserving named post-tonal constraints and explaining where it satisfies or violates them?

This question requires a different interface and a different evaluation vocabulary. Pitch-class set theory supplies normal order, prime form, and interval vectors (Forte, 1973; Straus, 2016); serial practice requires explicit $P/R/I/RI$ transformations and aggregate tracking; rhythmic and gestural intentions are expressed over score time rather than audio frames. The output must also be inspectable as notation. A fluent token sequence is therefore insufficient if it cannot be converted into a score or if its relation to the requested material remains opaque.

Data legality presents a second challenge. A large fraction of relevant post-1945 repertoire remains copyrighted, and a scraped corpus would be difficult to release or reproduce. We instead use a deterministic procedural corpus as a controlled testbed. The corpus is not intended to imitate a composer or establish historical style. Its purpose is to expose known condition–target relations so that constraint adherence, failure modes, and decoding trade-offs can be measured without distributing protected scores.

We study a condition-prefix Transformer that generates quantized event sequences and a neural-symbolic decoder that reranks sampled candidates using non-differentiable post-tonal diagnostics. The approach is deliberately modest at the neural level: it does not introduce a new attention architecture. Its contribution lies in connecting causal symbolic generation to pitch-class-set, serial, rhythmic, gestural, register, and notation-level checks, and in returning both a MusicXML sketch and a machine-readable explanation.

The corrected study addresses three research questions:

- **RQ1:** With one trained generator and fixed test conditions, how does four-candidate symbolic reranking change explicit constraint metrics relative to its aligned first candidate?
- **RQ2:** How does hiding pc-set, serial, rhythm, or gesture fields from the condition prefix affect adherence to the unchanged held-out targets?
- **RQ3:** Can generated fragments be exported as structurally valid MusicXML with the requested measures and voices and with inspectable analysis reports?

The main contributions are:

1. a legal and reproducible 24,000-fragment synthetic corpus protocol for score-level post-tonal conditions;
2. a compact condition-prefix/event representation and locally trainable Transformer conditioned on requested spans of 4–16 measures and 2–8 voices;

3. an inference-time candidate-reranking objective grounded in pc-set, serial, rhythmic, gestural, and range diagnostics; and

4. a provenance-gated evaluation and export layer that connects aggregate metrics to MusicXML examples and per-fragment JSON reports.

The resulting claims are intentionally bounded. The quantitative findings concern a synthetic constraint testbed, not stylistic authenticity. Expert judgments of compositional usefulness are planned with the supplied blind-rating package but are not inferred from automatic metrics. The completed experiment therefore supports claims about implemented symbolic control and notation export while leaving artistic value and transfer to independent scores unresolved.

2 Related Work

2.1 Computer-assisted composition and post-tonal representation

Computer-assisted composition has a longer history than data-driven music generation. OpenMusic represents musical objects and transformations in a visual programming environment, allowing a composer to formalize and revise compositional procedures without reducing the task to audio synthesis (Assayag et al., 1999). The music21 toolkit provides a complementary programmatic layer for symbolic score analysis, transformation, and interchange (Cuthbert and Ariza, 2010). These systems establish symbolic inspectability as a practical requirement for compositional software.

Constraint programming offers a second precedent for explicit musical control. Anders and Miranda (2011) survey systems that express theories of harmony, rhythm, form, and instrumentation as declarative constraints. This tradition treats a musical rule as an inspectable compositional object rather than as an implicit property of a corpus. Our decoder follows the same preference for explicit diagnostics, while using sampled neural candidates instead of solving a complete constraint-satisfaction problem.

Pitch-class set theory supplies equivalence under transposition and inversion, normal and prime forms, and interval-class vectors for non-tonal pitch organization (Forte, 1973; Straus, 2016). Post-tonal theory also treats ordered twelve-tone rows, $P/R/I/RI$ transformations, and aggregates as distinct from unordered set classes. The present system keeps that distinction in both its condition representation and its analysis report.

2.2 Neural symbolic music generation

Neural music generation spans different objectives, representations, and forms of control (Briot et al., 2020). The Transformer architecture provides a standard causal sequence model (Vaswani et al., 2017), and Music Transformer adapted self-attention to long symbolic performance sequences

(Huang et al., 2019). Later event representations made meter and beat structure more explicit for expressive pop-piano generation (Huang and Yang, 2020). These studies show that event sequences can support long musical outputs, but their principal corpora and evaluation tasks do not directly measure pitch-class-set adherence, serial transformations, or post-tonal gesture control.

Other symbolic models emphasize polyphonic coordination or latent structure. MuseGAN generates several instrumental tracks in a piano-roll representation and evaluates a rock-music accompaniment setting (Dong et al., 2018). MusicVAE uses a hierarchical recurrent decoder to model longer musical sequences in a continuous latent space (Roberts et al., 2018). These objectives are relevant to multi-part organization and variation, but neither formulation exposes the pitch-class-set and row-form conditions required by the present task.

Our model therefore uses the Transformer as an established sequence generator rather than claiming an architectural advance. Its event vocabulary is intentionally smaller than performance-oriented encodings because the output target is a quantized, editable score fragment. Conditions for pitch, serial order, rhythm, gesture, instrumentation, and span are written into the same symbolic sequence as a prefix.

2.3 Constraint-aware generation

Several systems make neural generation steerable. DeepBach permits positional constraints on notes, rhythms, and cadences during pseudo-Gibbs sampling (Hadjeres et al., 2017). Anticipation-RNN incorporates future unary constraints into sequence generation through a backward constraint network and a forward token network (Hadjeres and Nielsen, 2020). Outside purely neural sampling, MorpheuS uses optimization to preserve recurrent patterns and follow a target tension profile (Herremans and Chew, 2019). These methods motivate explicit control, but they address different musical domains and constraint types.

FIGARO is a closer precedent for interpretable conditional generation. It reconstructs symbolic music from learned and expert descriptions, including time-varying high-level features (von Rütte et al., 2023). Our study narrows the control vocabulary to post-tonal compositional materials and adds deterministic analysis after sampling. The resulting comparison asks whether sampling and scoring several candidates by pitch-class, serial, rhythmic, gestural, and range diagnostics changes adherence while the trained checkpoint and test requests remain fixed.

The approach studied here separates probabilistic generation from deterministic score analysis. A causal Transformer samples several candidates, after which post-tonal functions rerank them by pitch-class coverage, interval-vector distance, row order, aggregate completion, rhythmic profile, gesture features, and instrumental range. This design accepts the computational cost of multiple candidates in exchange for diagnostics that can be shown to a composer and saved with the score.

2.4 Position of the present study

The study connects three established practices: formal symbolic manipulation from computer-assisted composition, event-based neural sequence generation, and constraint-aware selection. Its narrower contribution is a workflow conditioned on 4–16-measure post-tonal sketch requests that returns both MusicXML and an explanation of constraint adherence. The legal synthetic corpus makes the condition–target relation reproducible. The experiment isolates candidate reranking by holding the checkpoint and first sampled candidate fixed, then tests condition fields through separately trained prefix-removal models evaluated against unchanged targets.

3 Methodology

3.1 Task formulation

The system generates a quantized symbolic score fragment x from a condition bundle c. The bundle may contain a pitch-class set S, a six-entry interval vector $v(S)$, a twelve-tone row r, a serial form $f \in \{P_n, R_n, I_n, RI_n\}$, a rhythmic profile, a gesture label, an instrument, a voice count, and a measure count. The requested output contains 4–16 measures and 2–8 parts and is returned as event tokens, MusicXML, and a score-level analysis report. Figure 1 summarizes this score-level path. No waveform or audio representation is used.

The conditional language model factorizes

$$p_\theta(x \mid c) = \prod_{t=1}^{T} p_\theta(x_t \mid c, x_{<t}), \tag{1}$$

where x_t is a score-event token. The research objective is to combine next-token modeling with explicit, inspectable post-tonal constraints.

3.2 Legal procedural corpus

Because much post-1945 art-music repertoire remains copyrighted, the bundled corpus is generated procedurally rather than scraped from published scores. Each sample specifies 4–16 measures, 2–8 voices, one instrument label, one of seven rhythmic profiles, and one of seven gesture labels. Non-serial samples receive a pitch-class set and interval vector. Serial samples instead receive a valid twelve-tone row and one of 48 transposed $P/R/I/RI$ forms. This separation prevents an impossible target in which one passage is simultaneously required to remain in a small set and complete the chromatic aggregate.

The rule generator creates a rhythmic skeleton for each voice, applies gesture-dependent onset, duration, rest, and register operations, and selects pitches inside the requested instrument range. Time is quantized to 0.25 quarterLength. The random density curve realized in each target is saved for held-out evaluation, but it is removed from model-visible condition metadata and is not used by candidate reranking.

3.3 Condition and event representation

Conditions are serialized as a prefix beginning with `BOS` and ending with `SEP`. The prefix can include pitch-class, interval-vector, row-pitch, row-form, rhythmic-profile, gesture, voice-count, measure-count, and instrument tokens. Explicit `NO_PCSET`, `NO_ROW`, `NO_PROFILE`, and `NO_GESTURE` symbols identify absent conditions.

The score body contains `BAR`, `TIME_SHIFT`, `VOICE`, `PITCH`, `OCTAVE`, `DUR`, and `REST` events followed by `EOS`. A finite-state grammar constrains decoding order, requested voice indices, instrument range, and requested bar count. The parser clamps decoded events to the requested span.

Many complete fragments exceed the 256-token model context. Training therefore uses windows that preserve the conditions instead of silently truncating the sequence. Every window repeats the complete condition prefix and retains preceding score context when capacity permits. Loss is applied only to the current score-body target segment. A ten-epoch coverage cycle partitions the target windows so that every saved score-body token contributes once per cycle. Validation and teacher-forced evaluation enumerate all windows. Reported token accuracy and cross-entropy are body-token diagnostics, not measures of compositional quality.

3.4 Conditional Transformer

The generator is a causal Transformer (Vaswani et al., 2017) with six layers, hidden dimension 384, six attention heads, GELU feed-forward blocks of width $4d$, learned positional embeddings, pre-normalization, and dropout 0.1. The training objective is teacher-forced negative log likelihood over the unmasked score-body labels:

$$\mathcal{L}_{\mathrm{NLL}}(\theta) = -\sum_{t\in\mathcal{T}} \log p_\theta(z_t \mid z_{<t}), \tag{2}$$

where $\mathcal{T}$ denotes the current target window. No pretrained language model or external score corpus is used.

3.5 Constraint-guided candidate reranking

At inference, the proposed decoder samples K grammar-constrained candidates and analyzes each at score level. Candidate selection is

$$x^* = \arg\min_{x\in\mathcal{C}_K} \sum_j \lambda_j m_j(x, c), \tag{3}$$

where m_j includes missing pc-set coverage, off-set pitch classes, interval-vector distance, cyclic row-order error, incomplete strict serial forms, incomplete serial aggregates, rhythmic-profile distance,

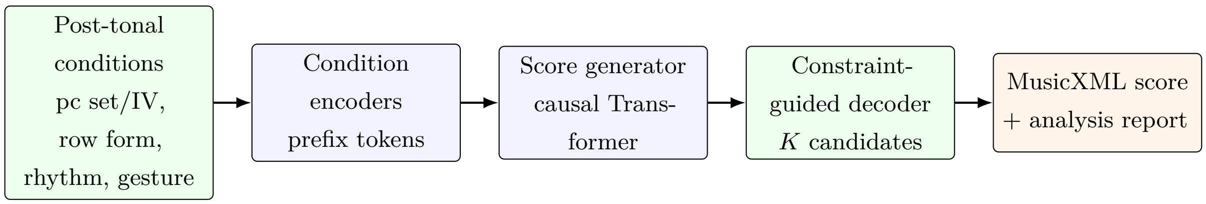

Figure 1: Corrected inference path. Post-tonal conditions are serialized by condition encoders, consumed by the causal score generator, and evaluated by a constraint-guided decoder before MusicXML and an analysis report are exported. A legal procedural generator supplies the condition–score pairs used for training.

gesture inconsistency, instrument-range violations, insufficient content span, and missing requested voices. Pitch-class terms are inactive without a pc-set target, and serial-form and aggregate terms are inactive without a row target. The held-out stochastic density curve is excluded from this objective. These non-differentiable terms are used only for candidate selection and are not presented as training losses.

The primary controlled contrast uses one shared trained generator: $K = 1$ is the vanilla decoder and $K = 4$ is the guided decoder. A deterministic random-number stream is assigned to each test condition, making the first $K = 4$ candidate identical to the corresponding $K = 1$ output. Condition-removal models use the same corpus samples and target events while modifying only model-visible prefix fields. Evaluation retains the original hidden targets.

3.6 Analysis and MusicXML export

Each selected fragment receives pc-set coverage, pc-set precision and Jaccard similarity, generated interval vector, interval-vector distance, cyclic row-order accuracy, strict complete-form accuracy, serial aggregate completion, density curve, rhythmic-profile distance, held-out density-curve error, gesture features, gesture consistency, register spread, range violations, content-span adherence, and requested-voice adherence. Strict serial-form accuracy counts complete non-overlapping twelve-note blocks and is independent of cyclic row-order accuracy.

MusicXML export creates the requested number of score parts and measures, pads empty trailing measures, and removes events outside the requested span (W3C Music Notation Community Group, 2021). Success is recorded separately for structural parsing, requested measure count, and requested part count.

4 Experimental Setup

4.1 Corpus and split

The procedural corpus contains 24,000 fragments with explicit saved membership: 20,000 training, 2,000 validation, and 2,000 test samples. Generation uses seed 42, 4–16 measures, 2–8 voices, six instrument labels, seven rhythmic profiles, and seven gesture labels. The split summary records `smoke=false`. CPU smoke samples are stored separately and do not enter formal metrics.

4.2 Training protocol

All formal neural configurations use a six-layer Transformer with hidden size 384, six heads, feed-forward size 1,536, dropout 0.1, and context length 256. Optimization uses AdamW with learning rate 3×10^{-4}, weight decay 0.01, physical batch size 16, gradient clipping at norm 1.0, fp16 autocast, and zero data-loader workers on Windows. Training is limited to 60 epochs with early-stopping patience 10. If CUDA memory is insufficient, the runner retries batch size 8 with two accumulation steps and then batch size 4 with four accumulation steps without reducing the corpus.

Each epoch writes a resumable last checkpoint containing model, optimizer, scaler, random-number states, epoch history, and corpus/configuration hashes. The best validation checkpoint is stored separately. Formal evaluation is permitted only when the checkpoint and completed training summary match the current configuration and processed-corpus hashes.

4.3 Hardware and software gate

The formal run used Windows 11, Python 3.11.9, PyTorch 2.5.1+cu121, CUDA, and an NVIDIA GeForce RTX 4060 Ti with 16GB VRAM. Peak measured process RAM across completed training runs was 4.637GiB, and peak allocated CUDA memory was 0.667GiB. No out-of-memory adjustment was required.

4.4 Compared configurations

The full runner evaluated thirteen named configurations: an independent procedural rule reference, a shared neural generator with single-candidate decoding, the same generator with four-candidate constraint-guided decoding, two no-constraint controls, four single-condition removals, and four focused-condition models. All neural condition ablations use the same saved corpus and split. The rule reference generates a fresh deterministic realization for each test condition and never reads the stored target event list.

The proposed and vanilla rows share one trained conditional generator, but the comparison jointly changes the candidate budget from one to four and activates symbolic candidate scoring.

Ablation models were trained separately with the specified condition-prefix fields hidden and used single-candidate decoding, matching the vanilla decoding budget. Metrics for ablations were computed against the unchanged original targets. The matrix contains twelve neural checkpoint paths but eleven distinct neural optimization runs because the vanilla and proposed rows share identical model parameters. Every neural configuration reached the 60-epoch limit; recorded best epochs ranged from 56 to 60.

4.5 Evaluation and export

Teacher-forced token metrics enumerate all score-body windows in the 2,000-item test split. Generation metrics use 2,000 test conditions per configuration. Twenty examples per evaluated configuration were attempted for MusicXML export, and the final expert package accepted only files that parsed structurally and matched requested measure and voice counts. All thirteen metric rows, all required checkpoint paths, and all 260 attempted export records passed the runner's final gate. Legacy v2 checkpoints, smoke outputs, and historical multi-seed development runs are excluded from the reported results.

5 Evaluation Metrics

Body-token diagnostics. Token accuracy is the fraction of unmasked score-body targets assigned the maximum logit. Cross-entropy is accumulated over the same target tokens across all explicit windows. Condition-prefix and overlap-context positions are masked.

Pitch-class-set measures. For generated support $P(x)$ and nonempty target set S, coverage is $|P(x) \cap S|/|S|$, precision is $|P(x) \cap S|/|P(x)|$, and Jaccard similarity is $|P(x) \cap S|/|P(x) \cup S|$. Interval-vector distance is the L_1 distance between generated and target six-entry interval-class vectors. These metrics are not applicable when no pc-set target exists.

Serial measures. Cyclic row-order accuracy compares generated pitch classes with repeated positions of the requested transformed row. Strict serial-transformation accuracy counts complete non-overlapping twelve-note blocks that exactly equal the requested form. Aggregate completion is the proportion of the twelve pitch classes present and is averaged only over row-conditioned samples.

Rhythmic measures. The density curve counts note onsets per measure. Rhythmic-profile distance compares the generated curve, normalized by requested voice count, with a deterministic profile template available to decoding. Density-curve error compares the generated curve with the stochastic target curve saved during corpus generation. The latter is held out from model conditions and reranking.

Gesture measures. Gesture features include onset dispersion, register spread, note density normalized by requested voices, mean duration, and per-voice rest coverage. Rest coverage is computed from interval unions and is bounded in $[0, 1]$. Feature closeness is averaged into a heuristic gesture consistency score. This score is explainable but is not a perceptual rating.

Range, span, voice, and notation checks. Range-violation rate is the proportion of pitched events outside the requested instrument range. Content-span adherence is the realized event end time divided by the requested duration, capped at one. Voice adherence is the fraction of requested voice indices containing generated material. MusicXML evaluation separately reports structural parse success, requested measure-count adherence, and requested part-count adherence.

Applicability and aggregation. Every aggregate is computed only over samples for which the corresponding target exists. Missing-target values are written as not applicable rather than zero or one. Formal CSV rows contain all required metric columns and exactly 2,000 evaluated conditions. All thirteen rows used below passed the provenance and completion gates before their canonical CSV and LaTeX tables were promoted.

6 Results

6.1 Full-run completion

The formal corpus comprised 20,000 training, 2,000 validation, and 2,000 test fragments, with `smoke=false`. Training and evaluation ran under Python 3.11.9 and PyTorch 2.5.1+cu121 with CUDA available on an NVIDIA GeForce RTX 4060 Ti 16GB. Thirteen configurations were evaluated on the same 2,000-condition test split. These comprised one rule reference and twelve neural rows, with the proposed $K = 4$ and vanilla $K = 1$ rows sharing one checkpoint. Canonical metrics are stored in `results/project2_metrics.csv` and `results/project2_constraints.csv`; per-example records are stored in `results/project2_generation_examples.json`.

6.2 Guided candidate selection with a shared generator

Increasing the candidate budget from one to four and applying symbolic scoring moved every plotted selection endpoint in its preferred direction relative to the aligned first candidate (Table 1; Figure 2a). Pc-set coverage increased from 0.9963 to 0.9999 while pc-set precision increased from 0.9937 to 1.0000. Interval-vector distance decreased from 0.2390 to 0.0045. Cyclic row-order accuracy increased from 0.1581 to 0.2411, and aggregate completion increased from 0.9949 to 0.9969.

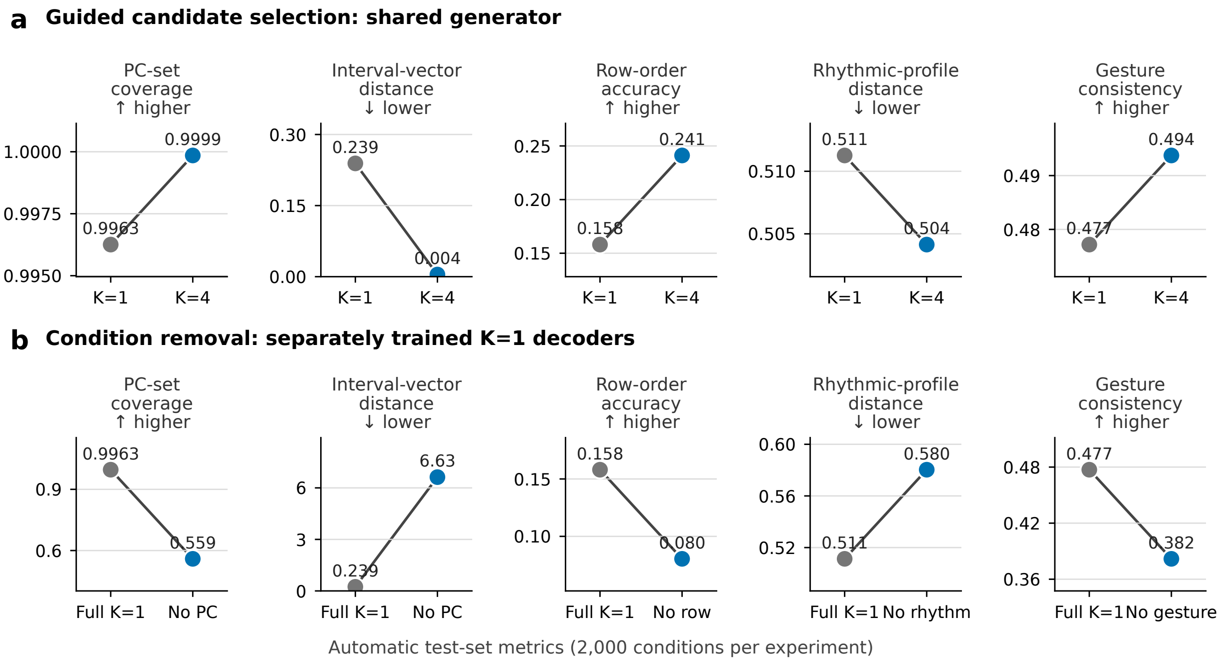

Figure 2: Automatic constraint effects on the fixed 2,000-condition test split. (a) The $K = 1$ and $K = 4$ points use one shared generator and aligned random streams; $K = 4$ samples four candidates and applies symbolic scoring. (b) The full-prefix and condition-removal models were trained separately, all use $K = 1$ decoding, and are evaluated against unchanged targets. Arrows state the preferred metric direction. These are descriptive single-seed automatic endpoints, not perceptual ratings or inferential estimates.

Rhythmic-profile distance decreased from 0.5113 to 0.5041, and gesture consistency increased from 0.4772 to 0.4937. The held-out density-curve error, which was unavailable to candidate scoring, decreased descriptively from 12.3501 to 12.2353.

Teacher-forced token accuracy was identical at 0.6686 because both rows use the same checkpoint. Strict serial-transformation accuracy remained 0.0000 for both decoders. Thus, reranking improved partial cyclic row adherence but did not yield exact complete twelve-note forms. Mean content-span ratio increased from 0.5024 to 0.5195, leaving almost half of the requested temporal extent without generated content on average even though export padded the requested measure count.

6.3 Condition fields and procedural reference

Removing a condition field from a separately trained $K = 1$ model degraded its matching endpoint relative to the full-prefix $K = 1$ model (Figure 2b). Without the pc-set field, coverage fell from 0.9963 to 0.5593 and interval-vector distance rose from 0.2390 to 6.6267. Without the

Experiment	Token acc.	PC cov.	PC prec.	IV dist.	Row acc.	Aggregate	Form acc.	Rhythm dist.	Density err.	Gesture	Range viol.	Span	Content voices	XML parse	XML measures
Shared generator, K=4 guided	0.6686	0.9999	1.0000	0.0045	0.2411	0.9969	0.0000	0.5041	12.2353	0.4937	0.0000	0.5195	1.0000	1.0000	1.0000
Shared generator, K=1	0.6686	0.9963	0.9937	0.2390	0.1581	0.9949	0.0000	0.5113	12.3501	0.4772	0.0000	0.5024	1.0000	1.0000	1.0000
Rule reference	–	0.9988	1.0000	0.0260	1.0000	0.9958	0.9819	0.3021	2.9263	0.5683	0.0000	0.9986	1.0000	1.0000	1.0000

Table 1: Automatic results on the fixed 2,000-condition procedural test split. The guided and single-candidate rows share one checkpoint; the rule reference implements constraints directly. Row and serial-form accuracy are averaged over row-conditioned samples. XML parse and measure adherence are measured over 20 attempted exports per configuration; XML part-count adherence is 1.0000 for all three rows.

serial field, row-order accuracy fell from 0.1581 to 0.0804. Without the rhythm field, rhythmic-profile distance rose from 0.5113 to 0.5803 and held-out density error rose from 12.3501 to 13.6056. Without the gesture field, gesture consistency fell from 0.4772 to 0.3818. These contrasts show that the learned generator used the corresponding prefix fields on the procedural distribution; they do not establish generalization to human-authored repertoire.

The rule reference directly implements the requested procedures and therefore serves as a construction reference rather than a learned baseline. It reached row-order accuracy 1.0000 and strict serial-form accuracy 0.9819, compared with 0.2411 and 0.0000 for the proposed neural decoder. It also produced a lower rhythmic-profile distance of 0.3021 and a lower density error of 2.9263. These gaps identify exact serial ordering and rhythmic realization as unresolved neural-generation problems.

6.4 MusicXML and inspectable examples

All 260 attempted exports across the thirteen configurations parsed as MusicXML and matched the requested measure and part counts. Range-violation rate was 0.0000 for every aggregate row, and requested-voice adherence was 1.0000 except for negligible rounding in two focused ablations. The proposed-model expert package contains twenty MusicXML files, twenty paired JSON reports, a manifest, and blank blind-rating forms under `expert_eval/project2/`.

Figure 3 shows one proposed-model example from that package. Its conditions were pc set $\{0, 1, 2, 5, 6\}$, a sustained rhythmic profile, registral expansion, four voices, and four measures. Its automatic report records pc-set coverage and precision of 1.0000, interval-vector distance 0.0000, rhythmic-profile distance 0.3125, gesture consistency 0.7565, content-span ratio 1.0000, and no range violation. The example is provided for notation-level inspection and is not evidence of artistic quality.

Experiment	Token acc.	PC cov.	PC prec.	IV dist.	Row acc.	Aggregate	Form acc.	Rhythm dist.	Density err.	Gesture	Range viol.	Span	Content voices	XML parse	XML measures
No constraints (seed 42)	0.6668	0.7097	0.4970	24.4978	0.0838	0.5942	0.0000	0.5885	13.9216	0.4014	0.0000	0.4664	1.0000	1.0000	1.0000
No pc-set token	0.6691	0.5593	0.5615	6.6267	0.1436	0.9951	0.0000	0.5119	12.3192	0.4754	0.0000	0.5043	0.9999	1.0000	1.0000
No row token	0.6688	0.9963	0.9928	0.2704	0.0804	0.9972	0.0000	0.5114	12.3576	0.4774	0.0000	0.5009	1.0000	1.0000	1.0000
No rhythm token	0.6673	0.9971	0.9932	0.2202	0.1513	0.9981	0.0000	0.5803	13.6056	0.4785	0.0000	0.4745	1.0000	1.0000	1.0000
No gesture token	0.6686	0.9945	0.9961	0.2175	0.1559	0.9946	0.0000	0.5127	12.9772	0.3818	0.0000	0.4991	1.0000	1.0000	1.0000
Serial only	0.6663	0.5696	0.5642	6.6106	0.0874	0.9973	0.0000	0.5852	13.9426	0.3879	0.0000	0.4612	1.0000	1.0000	1.0000
PC-set only	0.6662	0.9959	0.9936	0.2435	0.0817	0.9974	0.0000	0.5936	14.0748	0.3984	0.0000	0.4515	1.0000	1.0000	1.0000
Rhythm only	0.6679	0.7513	0.4836	28.2265	0.0812	0.6697	0.0000	0.5088	12.9737	0.3831	0.0000	0.4863	1.0000	1.0000	1.0000
Gesture only	0.6677	0.7486	0.4833	28.2632	0.0838	0.6484	0.0000	0.5802	13.5792	0.4769	0.0000	0.4764	0.9999	1.0000	1.0000
No constraints (seed 53)	0.6676	0.7264	0.4932	25.5506	0.0870	0.6019	0.0000	0.5767	13.7871	0.3850	0.0000	0.5109	1.0000	1.0000	1.0000

Table 2: Single-candidate condition-prefix ablations and focused-condition models on the fixed procedural test split. Each model sees its configured prefix while evaluation retains the original held-out targets. Row and serial-form accuracy are averaged over row-conditioned samples. XML parse and measure adherence are measured over 20 attempted exports per configuration; XML part-count adherence is 1.0000 for all rows.

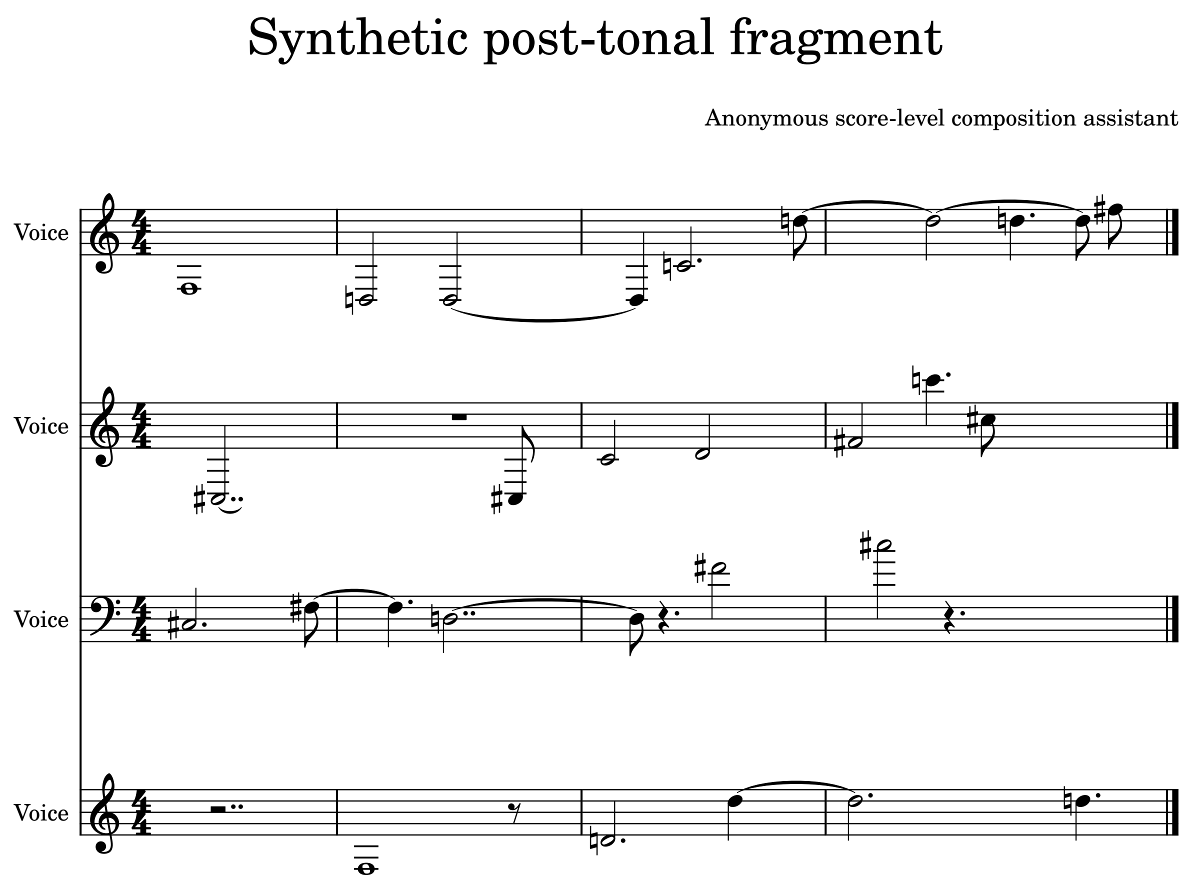

Figure 3: Representative proposed-model output `project2_20`, rendered from the exported MusicXML. The four requested parts and four measures are visible. Empty space, register, and durations are generated symbolic decisions; no human preference rating has been assigned to this example.

7 Discussion

The shared-checkpoint comparison answers RQ1 at the level of implemented diagnostics. Increasing the candidate budget and applying symbolic scoring improved pc-set, interval-vector, cyclic row-order, rhythmic-profile, and gesture endpoints without changing teacher-forced token accuracy. The contrast therefore concerns the combined guided-decoding procedure rather than a stronger trained language model or reranking alone. The held-out density curve also changed slightly in the preferred direction, although it was not available to the selector and no inferential uncertainty is reported.

The condition-removal experiments address RQ2 more directly. Each removal produced its largest interpretable loss in the associated control dimension. The pronounced pc-set and interval-vector changes indicate that pitch-class tokens do more than label a fixed generator distribution. Likewise, row, rhythm, and gesture fields contribute information that cannot be recovered fully from the remaining prefix. Because the corpus is procedural and each comparison uses one fixed seed, these effects should be understood as internal validity checks rather than evidence of broad musical generalization.

The serial results expose an important distinction between aggregate presence and ordered realization. The proposed decoder covered almost the full chromatic aggregate on serial targets, yet exact serial-transformation accuracy was zero and cyclic row-order accuracy was only 0.2411. Candidate reranking therefore selected fragments with better local or cyclic agreement without solving complete $P/R/I/RI$ realization. The procedural reference's near-perfect strict-form score confirms that the metric is attainable under direct construction and that the remaining gap lies in the learned generation path.

RQ3 is supported only at the structural notation level. Every attempted MusicXML file parsed and matched requested measures and parts, but mean content-span ratio for the proposed model was 0.5195. Measure-count adherence partly reflects explicit padding rather than continuous musical activity. The exports are consequently usable as editable sketches and audit objects, while publication-quality engraving and compositional usefulness remain questions for expert review.

More broadly, the results favor a division of labor between probabilistic generation and explicit symbolic analysis. The neural model supplies variation, while deterministic theory functions expose and rank constraint adherence. The rule reference remains stronger for exact serial and rhythmic realization, suggesting that future assistants may benefit from hybrid decoding in which selected constraints are constructed exactly and neural generation operates within the remaining degrees of freedom.

8 Limitations

The procedural corpus is legal and reproducible but represents one designed distribution rather than the breadth of contemporary compositional practice. A model can learn the generator's regularities without transferring to independent human-authored scores. Validation on legally supplied MusicXML and blind ratings by composers or post-tonal analysts have not yet been conducted.

The formal matrix is a single-seed descriptive experiment. It does not estimate variation across training initializations, corpus seeds, or candidate-sampling streams, and no inferential significance test is reported. Most constraint endpoints also contribute to candidate selection, so their $K = 4$ improvements measure the implemented selection mechanism rather than an independent perceptual criterion. The held-out density curve provides a partial exception but changed only modestly.

Automatic metrics reduce complex musical judgments to inspectable features. Pc-set coverage is recall-like, interval vectors ignore ordering and register, cyclic row accuracy depends on serialized note order, and gesture consistency uses hand-specified feature targets. The zero strict serial-form accuracy of all neural configurations shows that high aggregate completion must not be read as successful twelve-tone realization.

The model also underfills requested spans. Although MusicXML measure and part counts were exact, the proposed model's mean content-span ratio was 0.5195 because empty trailing time was padded during export. Structural validity therefore does not establish sustained material across the requested duration, publication-quality notation, or practical usefulness in a composer's workflow.

The event vocabulary uses a 4/4 sixteenth-note grid and omits tuplets, microtonality, changing meters, extended techniques, detailed articulation, evaluated dynamics, and engraving layout. One instrument label is repeated across parts, so mixed-ensemble orchestration is outside the current interface. The 256-token context, fixed four-candidate budget, fixed penalty weights, and absence of weight or temperature sensitivity studies further limit the present conclusions.

9 Reproducibility Statement

All bundled data are generated from source code and fixed seeds. No copyrighted post-1945 score is downloaded or required. The processed corpus is stored as `data/processed/project2_v3_main.pt`; its explicit 20,000/2,000/2,000 split and SHA-256 digest are recorded in `results/project2_full_split_summary.json`. The split records seed 42, `smoke=false`, and zero discarded score-body tokens under the coverage-cycle training strategy.

On Windows, `scripts/smoke_project2.ps1` runs the isolated CPU verification workflow, while `scripts/run_project2_full_local.ps1-Resume` runs and audits the CUDA matrix. The completed formal environment used Python 3.11.9, PyTorch 2.5.1+cu121, CUDA, and an NVIDIA GeForce RTX 4060 Ti. Peak measured process RAM was 4.637GiB and peak allocated CUDA memory was 0.667GiB. No out-of-memory adjustment occurred.

Every neural training directory contains a best checkpoint, a resumable last checkpoint, and a summary with optimizer state, scaler state, random-number state, epoch history, best epoch, configuration hash, corpus hash, elapsed time, and resource peaks. All neural rows reached 60 epochs. A single Windows CUDA-driver startup fault occurred before `pcset_only` training; the logged health check and resumed run completed that configuration without changing its data or settings. The incident is retained in `results/project2_full_run_incidents.json`.

Canonical aggregate outputs are `results/project2_metrics.csv`, `results/project2_constraints.csv`, and `results/project2_generation_examples.json`. The complete command history, checkpoint hashes, completed configurations, environment audit, and artifact gates are in `results/project2_full_run_report.md`. Manuscript tables are generated from the canonical CSV files, and Figure 2 has a saved source CSV and build script under `paper/figures/`. The twenty-example expert package and its manifest are under `expert_eval/project2/`.

The experiment-completion revision is recorded in the full-run report; the present manuscript revision adds no new experimental values. The source repository location is given on the title page for the identified manuscript. Because the repository is private during review and large tensors and checkpoints may be excluded from version control, an archival release must include code, configurations, hashes, metric files, tables, representative MusicXML, and regeneration instructions.

10 Conclusion

This study integrates a legal procedural corpus, condition-preserving Transformer training, pitch-class-set and serial analysis, grammar-constrained candidate generation, MusicXML export, and per-fragment reports in one score-level composition-assistance pipeline. On the fixed synthetic test set, increasing the candidate budget from one to four and applying symbolic scoring improved each targeted automatic endpoint relative to an aligned single candidate. Separately, removing pc-set, serial, rhythm, or gesture fields from single-candidate models degraded the corresponding control metric.

The same results define the method's present boundary. Neural generation did not realize any complete requested serial form exactly, and its average material occupied only about half of the requested temporal span despite structurally correct MusicXML. The system is therefore best understood as an inspectable sketch generator rather than a replacement for formal rule construction or compositional judgment. Blind expert evaluation, independent legal-score validation, and replicated training remain necessary before making claims about artistic usefulness or transfer beyond the procedural testbed.

Data and Code Availability Statement

The study uses only reproducible rule-generated symbolic data; no copyrighted contemporary score corpus is required. Source code, configurations, corpus-generation instructions, result CSV/JSON files, manuscript tables, and representative MusicXML examples are maintained with the project. The source repository is maintained at https://github.com/niuyupeng/yinyue2. At the time of writing it is private; an archival release will follow the journal's review and data-sharing policy. The processed corpus and neural checkpoints are large derived artifacts and can be regenerated from the fixed configuration, seed, and hashes reported in the repository.

An archival release should preserve the experiment-completion revision `47f8e3c`, the canonical metric files, and the twenty-example expert package.

Declarations

No human participants or copyrighted post-1945 scores were used in the reported experiment. Blind expert ratings and independent external MusicXML validation have not been conducted and are not represented as evidence. Funding, competing-interest, author-contribution, and AI-assistance disclosures require author confirmation before the identified manuscript is submitted. In the anonymous version, identifying declarations are withheld for review and must be supplied through the journal's required disclosure fields.

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